

Evaluating the *Behavior* of Large Language Models: Bias, Harm, and Red-Teaming — A State of the Art (2024–2026)

Claude

Anthropic

A research essay on safety and alignment evaluation · 2026-07-08

ABSTRACT. Measuring what a large language model (LLM) *knows* is a largely solved engineering problem; measuring how it *behaves* under adversarial, ambiguous, or emotionally loaded conditions is not. This essay reviews the 2024–2026 state of the art in evaluating LLM behavior along two axes that dominate the field — **social bias** (dialectal, gender, racial, socioeconomic, intersectional) and **harmful behavior** elicited through **red-teaming** (jailbreaks, manipulation, multi-turn escalation). We survey the principal benchmarks and datasets, the methodologies practitioners actually use (human vs. automated red-teaming; single- vs. multi-turn; static vs. dynamic evaluation), the metrics and their known failure modes (Attack Success Rate, severity, refusal rate, LLM-as-judge), and the commercial and open-source tooling landscape. We are deliberately critical: a recurring theme is that the instruments we use to certify safety are themselves poorly validated. We close with a concrete, resource-aware methodological proposal and a set of open questions suitable for an original research project. All cited primary sources are archived in `state_of_art/`.

1. Context and problem

1.1 Capability is not behavior

Most public discourse about LLM quality is about **capability**: accuracy on MMLU, code that compiles, math that checks out. Capability evaluation has a comfortable property — a *ground truth*. There is a correct answer, and the model either produces it or does not.

Behavior evaluation has no such luxury. The questions are: *Will the model insult a user? Will it comply with a request to synthesize a nerve agent if asked cleverly enough? Does it quietly treat a speaker of African American English as less employable? Does it flatter, deceive, or manipulate?* These are properties of the model’s disposition under a distribution of *contexts and interlocutors*, not of a single input–output pair. Three structural difficulties follow:

1. **No canonical ground truth.** “Harmful,” “biased,” and “toxic” are contested, context-dependent social constructs. Whether a response is a policy violation depends on the deployer’s policy, the jurisdiction, and the conversational frame.

2. **The evaluation surface is adversarial and open-ended.** A capability benchmark can be a fixed test set; a behavior benchmark is chasing an adversary who only needs *one* working attack path while the defender must cover all of them.
3. **Behavior is emergent over interaction.** A model can be perfectly safe turn-by-turn and unsafe over a ten-turn conversation — a fact that, as we will see, breaks most static benchmarks.

1.2 What “harmful behavior,” bias, and misalignment mean here

- **Harmful behavior** — outputs that cause or materially facilitate harm: instructions for weapons or cyber-offense, targeted harassment, self-harm encouragement, non-consensual content, or fraud. In practice the field operationalizes this as *policy-violating output produced in response to a request the model was intended to refuse*.
- **Bias** — systematic, unjustified differences in how the model treats or represents social groups. This spans *representational* harms (stereotyping, demeaning associations) and *allocational* harms (differential recommendations about jobs, credit, criminality). Bias can be **overt** (explicit stereotypes) or **covert** (surfacing only through indirect cues such as dialect).
- **Misalignment** — a broader failure in which the model optimizes for something other than the operator’s or user’s genuine intent: sycophancy, deception, specification gaming, or pursuing instrumentally useful but undesired sub-goals. Bias and harmful behavior are observable *symptoms*; misalignment is the underlying disease.

The crucial methodological point: because there is no ground truth, **every behavioral number is only as trustworthy as the judgment procedure that produced it** — a theme that recurs in Sections 3–5.

2. State of the art in research (2024–2026)

2.1 Benchmarks for measuring bias

The classic quartet remains the shared vocabulary of the field: **BBQ** (Parrish et al., 2022; 2110.08193), a QA benchmark over nine U.S. social axes that cleverly separates behavior in *ambiguous* versus *disambiguated* contexts; **BOLD** (Dhamala et al., 2021; 2101.11718) for open-ended generation; **HolisticBias** (Smith et al., 2022; 2205.09209), built participatorily with ~600 descriptors across 13 axes; and the minimal-pair probes **StereoSet** (2004.09456) and **CrowS-Pairs** (2010.00133). These are indispensable but carry well-known limitations: they are overwhelmingly **U.S.-English-centric**, **template-based** (limited naturalness), and can only detect *pre-catalogued* biases. StereoSet and CrowS-Pairs in particular were sharply criticized by Blodgett et al. (2021) for data-quality and construct-validity problems, and both were designed for masked LMs — mapping them onto modern autoregressive, instruction-tuned models is awkward.

The most consequential recent bias finding is **Hofmann et al., *Nature* 2024** (2403.00742). Using *Matched Guise Probing* — comparing intent-equivalent African American English (AAE) and Standard American English texts — they show that LLMs harbor **covert** stereotypes about AAE speakers that are *more negative than any human stereotype about African Americans ever experimentally recorded*, assigning them less prestigious jobs, more convictions, and even more death sentences. Critically, the models’ *overt* statements about Black people are positive, and **human-preference alignment (RLHF)** **widened the gap** between overt and covert bias rather than closing it. This is the strongest evidence to date that surface-level “de-biasing” can mask rather than remove bias — and that dialect is a potent, under-tested attack surface. Supporting work includes **Multi-VALUE** (2212.08011), a rule-based translator across 50 English dialects that reveals large performance drops on non-standard dialects.

Socioeconomic and intersectional bias, historically neglected, gained traction in 2024: *Born With a Silver Spoon?* (2403.14633) introduces the SilverSpoon scenario dataset, and Arzaghi et al. (AIES 2024; 2405.18662) build a one-million-sentence corpus showing SES bias is **amplified under intersectionality** — models infer multiple demographic attributes from a name and compound the bias. The standard synthesis is **Gallegos et al., *Computational Linguistics* 2024** (2309.00770), whose taxonomy of bias definitions, metrics, and mitigations, with an accompanying benchmark repository, is the field’s de facto reference.

2.2 Benchmarks and methods for harmful behavior / red-teaming

The reference framework is **HarmBench** (Mazeika et al., ICML 2024; 2402.04249), which standardized the comparison of 18 attacks against 33 models under a unified Attack Success Rate (ASR) metric. **JailbreakBench** (2404.01318) adds a live leaderboard and an open repository of reproducible jailbreak artifacts; **SORRY-Bench** (2406.14598) contributes a fine-grained 45-topic refusal taxonomy. A pivotal *critical* contribution is **StrongREJECT** (2402.10260), which showed that many “successful” jailbreaks in earlier work actually *degrade the model’s capability* — the model emits garbage that a naive classifier scores as compliance. StrongREJECT therefore argues a large fraction of headline ASR numbers were inflated.

On the **attack** side, the lineage runs from white-box gradient attacks — **GCG** (Zou et al., 2023; 2307.15043), which introduced AdvBench and transferable adversarial suffixes — to efficient black-box, LLM-driven attackers: **PAIR** (2310.08419, jailbreak in <20 queries), **TAP** (2312.02119, tree-of-attacks, >80% ASR on GPT-4-class models), and **Rainbow Teaming** (2402.16822), which casts adversarial prompt generation as quality-diversity search.

The **multi-turn** results of 2024 are, arguably, the most important development for practitioners. Microsoft’s **Crescendo** (2404.01833) starts from a benign question and escalates gradually, steering the model with its own prior replies to reach high ASR across ChatGPT, Gemini, and Llama models. Scale AI’s **Multi-turn Human Jailbreaks (MHJ)** study (2408.15221) is the clincher: human red-teamers achieved **>70% ASR on HarmBench against defenses that report single-digit ASR under automated**

single-turn attacks, and even recovered “unlearned” dual-use knowledge. Anthropic’s **Many-shot Jailbreaking** (Anthropic research page; NeurIPS 2024 proceedings) shows effectiveness scaling as a *power law* in the number of in-context faux-compliance examples — a vulnerability created by the very long context windows that make models useful.

2.3 Recent findings and admitted research gaps

Three findings recur across the 2024–2026 literature:

1. **Frontier models remain non-robust to adaptive attacks.** Andriushchenko et al. (ICLR 2025; 2404.02151) reach **~100% ASR** on many aligned models with simple, per-model adaptive attacks, and crucially show ASR is *highly sensitive to attack budget, judge choice, and the definition of a successful target string* — i.e., it is not a stable single number.
2. **Single-turn evaluation systematically understates deployed risk** (MHJ, Crescendo, M2S 2503.04856). Guardrails tuned on single-turn English collapse under multi-turn and cross-lingual pressure (2504.03174).
3. **Alignment can hide rather than remove problems** (Hofmann et al.’s covert/overt gap).

The gaps the authors themselves admit are strikingly consistent: (a) **no agreed definition or taxonomy of “harm”** (Ganguli et al., 2209.07858); (b) coverage is overwhelmingly **English, text-only, single-turn**, leaving multilingual, multimodal, and agentic settings under-benchmarked; (c) **evaluations are lower bounds** — they can demonstrate a capability but never prove its absence (Phuong et al., 2403.13793; 2412.08653); and (d) the **judges and metrics are themselves unvalidated** (Section 3.3).

3. Methodologies: how this is done in practice

3.1 Human vs. automated red-teaming

Human red-teaming — skilled testers probing the model — remains the gold standard for *discovering* novel failure modes. The foundational study is Ganguli et al. (Anthropic, 2022; 2209.07858), which released ~39k human attacks and found RLHF models get harder to attack with scale. Its admitted weaknesses define the trade-off: human red-teaming is **expensive, slow, hard to reproduce, dependent on the tester’s identity and creativity**, and carries a real **psychological cost** to red-teamers exposed to disturbing content.

Automated red-teaming uses one model to attack another. The seminal work is Perez et al. (DeepMind, 2022; 2202.03286), “Red Teaming Language Models with Language Models.” Automated methods scale and reproduce cheaply but suffer **mode collapse** — they rediscover a handful of attack templates — which is why *diversity-seeking* methods like Curiosity-driven red-teaming (2402.19464) and Rainbow

Teaming exist. In practice, mature programs (DeepMind’s holistic evaluations, 2404.14068; frontier dangerous-capability batteries, 2403.13793) **combine both**: automated methods for breadth and regression, humans for depth and novelty.

3.2 Single-turn vs. multi-turn; static vs. dynamic

- **Single-turn / static benchmarks** (HarmBench, JailbreakBench, BBQ) are cheap, comparable, and reproducible — and systematically optimistic. They are best understood as *regression tests*, not risk estimates.
- **Multi-turn / dynamic evaluation** (Crescendo, MHJ) mirrors real usage and reveals far higher risk, but is expensive and harder to standardize. A survey of agent evaluation (2503.16416) reports that only ~2% of studies use methodology beyond fixed test cases — a striking imbalance given that deployed systems are agentic and conversational.
- **Static benchmarks additionally suffer contamination and saturation**: web-scraped test sets leak into training data (2406.04244), and leaderboards can be gamed. *The Leaderboard Illusion* (2504.20879) documents data-access asymmetry and undisclosed private testing on Chatbot Arena; a companion paper (2501.17858) demonstrates outright vote-rigging.

3.3 Metrics and their limitations

- **Attack Success Rate (ASR)** — the fraction of attempts that elicit a policy violation. It is the field’s workhorse and its most treacherous number: it depends on the attack set, the query budget, and *how a “success” is judged* (2404.02151).
- **Severity / harm scoring** — weighting violations by potential damage (e.g., a bioweapon recipe » mild profanity). Severity is more decision-relevant than raw ASR but is subjective and rarely standardized.
- **Refusal rate** — how often the model declines. It conflates *genuine safety* with *over-refusal* (a helpfulness cost), and is trivially gameable by a model that refuses everything.
- **LLM-as-a-judge** — using an LLM to score outputs is now ubiquitous because human grading doesn’t scale. It is also the field’s soft underbelly. The LLM-as-judge survey (2411.16594) catalogs position, verbosity, and self-preference biases and prompt-injection susceptibility; and *A Coin Flip for Safety* (2603.06594) finds LLM judges perform **near chance** when asked to decide whether an adversarial attack succeeded. **If the judge is a coin flip, every ASR built on it inherits that noise.** Any serious protocol must therefore report human-judge agreement and treat automated scores as estimates with error bars.

4. Tools available on the market

Tool	Type	What it measures	How it's used	Point-in-time vs. CI/CD
NVIDIA garak	OSS (Apache-2.0)	Jailbreaks, prompt injection, toxicity, leakage, hallucination (50+ probes)	Python CLI scanner, any API model	Mostly one-off; scriptable
Microsoft PyRIT	OSS (MIT)	Automated + human red-teaming; multi-turn (Crescendo, TAP, Skeleton Key)	Python framework + CoPyRIT GUI	Both; built for scale
promptfoo	OSS (MIT)	50+ attack plugins; OWASP LLM Top-10, MITRE ATLAS mappings	Declarative YAML, CLI/library	Strong CI/CD + one-off
Giskard	OSS + commercial Hub	Hallucination, injection, toxicity, robustness, bias; multi-turn agents	Python SDK; Hub adds scheduling	OSS one-off; Hub continuous
DeepEval	OSS (Apache-2.0)	50+ metrics incl. bias, toxicity, faithfulness (Pytest-style)	Python unit tests	Unit tests → native CI
Stanford HELM	OSS	Holistic: accuracy, robustness, fairness, bias, toxicity	Python framework, leaderboards	One-off benchmarking
UK AISI Inspect	OSS (MIT)	Frontier evals: tool use, multi-turn, model-graded; 200+ prebuilt evals	Python framework	Both; serious eval suites
Meta Purple Llama / Llama Guard	OSS weights	Input/output safety & prompt-injection classifiers; CyberSecEval	Guardrail models + benchmarks	Runtime + benchmarks
HuggingFace Evaluate	OSS (Apache-2.0)	Toxicity, Regard, HONEST (gender stereotype)	Python library	One-off / embeddable
Lakera Red / Guard	Commercial (Check Point)	Pre-prod attack simulation (Red); runtime	SaaS API	Red one-off; Guard continuous

Tool	Type	What it measures	How it's used	Point-in-time vs. CI/CD
		injection/jailbreak defense (Guard)		
Robust Intelligence	Commercial (Cisco AI Defense)	AI Validation (pre-deploy) + AI Firewall (runtime)	Enterprise platform	Both
Google Perspective API	Free API (<i>sunset 2026-12-31</i>)	Toxicity, identity attack, insult, threat	REST API	Per-request

Practical reading of the landscape: a common 2026 stack pairs **garak** + **PyRIT** (offensive discovery), **promptfoo** (CI/CD regression gating), and a **runtime guardrail** (Llama Guard / Lakera Guard / RI Firewall). Note the **consolidation wave** — promptfoo → OpenAI (2026), Lakera → Check Point (2025), Robust Intelligence → Cisco (2024) — so product names and URLs are migrating, and **Perspective API is being sunset**, so it should not anchor new long-term pipelines. No single tool covers bias *and* jailbreaks *and* runtime defense well; composition is mandatory.

5. Proposal: how one could conduct this research

The following is a concrete, viable design for an original study of *how current models behave*, scoped for a small team.

Models under test. Compare 4–6 models spanning the design space: two frontier proprietary (e.g., a Claude and a GPT-class model), two strong open-weight (e.g., a Llama and a Qwen/Mistral-class model), and one small local model as a baseline. Pin exact versions and sampling parameters — behavior drifts across releases.

Priority harm/bias categories. Resist the temptation to cover everything. Prioritize where the evidence says the risk is both high and under-measured: (1) **covert dialectal bias** (the highest-signal recent finding), (2) **intersectional socioeconomic bias**, (3) **multi-turn escalation** toward a small set of clearly policy-violating targets, and (4) **sycophancy/manipulation** as a misalignment probe.

Benchmark + red-team mix. Use existing benchmarks as a *reproducible floor* and add original red-teaming for *realism*: - *Bias*: BBQ + HolisticBias for breadth; reproduce Hofmann’s Matched Guise Probing for covert dialect bias; add SilverSpoon-style intersectional scenarios. - *Harm*: HarmBench/JailbreakBench single-turn as regression baseline; then a **custom multi-turn campaign** using PyRIT’s Crescendo and a small human red-team on the same target behaviors — precisely to measure the single-turn-vs-multi-turn gap on *current* models. - *Scoring*: use StrongREJECT’s autograder **and** a second independent judge, and **validate the judges against a human-labeled sample**, reporting agreement (Cohen’s κ). Never report an ASR without its judge’s error rate.

Documentation & reporting. Follow the emerging norms: publish a **model card / system card appendix** per model; release prompts and (where legal and safe) artifacts as in JailbreakBench; report ASR *with confidence intervals and query budget*, severity-weighted scores, refusal *and* over-refusal rates, and the covert–overt bias gap. Frame all numbers as **lower bounds**.

Resources, time, and limits. A minimal study is feasible in **~8–12 weeks** for 2–3 researchers: ~2 weeks harness/tooling (Inspect or PyRIT + promptfoo in CI), ~4–5 weeks running benchmarks and red-team campaigns, ~2 weeks judge validation and analysis, ~2 weeks writing. Budget for **API costs** (multi-turn attacks multiply token spend) and GPU time for open-weight models. **Ethical and technical constraints** are first-order, not footnotes: - **Dual-use.** Do not publish operational uplift (working weapon/cyber recipes). Report the *existence* of a vulnerability and its severity, not a runnable exploit; follow **coordinated disclosure** with the model vendor, per NIST AI 800-1 guidance. - **Human-subject care.** Red-teamers need content warnings, rotation, and psychological support. - **Construct validity.** Pre-register what each metric is *supposed* to measure and argue it is separable from raw capability (the “safetywashing” critique, 2407.21792). - **Contamination.** Prefer freshly authored or perturbed prompts; assume public benchmarks may be in training data.

6. Open questions for an original research project

1. **Can we build a trustworthy judge?** Given that LLM judges score adversarial robustness at near-chance (2603.06594), what ensemble, calibration, or human-in-the-loop protocol makes ASR a reliable number — and how do we certify it?
 2. **What is the true single-turn → multi-turn → agentic risk gradient** on 2026 frontier models, measured on identical target behaviors? MHJ established the gap on 2024 models; is it closing or widening as context windows and agentic scaffolds grow?
 3. **Does alignment hide bias?** Hofmann et al. found RLHF widened the covert–overt gap. Is this general across bias types and languages, and can an intervention shrink the *covert* gap without harming helpfulness?
 4. **How do we measure behavior without a ground truth we can trust?** What would a *construct-valid*, contamination-resistant, dynamically regenerated behavior benchmark look like in practice (per 2511.04703)?
 5. **Do benchmark numbers predict real-world harm?** There is little evidence linking ASR or bias scores to downstream incidents. What longitudinal or field-testing design (cf. NIST ARIA) would close that gap?
 6. **Multilingual and multimodal blind spots.** Guardrails degrade across languages (2504.03174); how large is the non-English, non-text harm surface on current models, and who is bearing that under-measured risk?
-

Selected references

(All arXiv PDFs archived in `state_of_art/`; see `state_of_art/SOURCES.md` for the full index, tool URLs, and governance documents.)

- Parrish et al. (2022). *BBQ: A Hand-Built Bias Benchmark for QA*. arXiv:2110.08193.
- Dhamala et al. (2021). *BOLD*. arXiv:2101.11718. · Smith et al. (2022). *HolisticBias*. arXiv:2205.09209.
- Nadeem et al. (2020). *StereoSet*. arXiv:2004.09456. · Nangia et al. (2020). *CrowS-Pairs*. arXiv:2010.00133.
- Hofmann, Kalluri, Jurafsky, King (2024). *Dialect prejudice predicts AI decisions...* *Nature* 633. arXiv:2403.00742.
- Ziems et al. (2023). *Multi-VALUE*. arXiv:2212.08011. · Singh et al. (2024). *Born With a Silver Spoon?* arXiv:2403.14633.
- Arzaghi et al. (2024). *Intrinsic Socioeconomic Biases in LLMs*. AIES. arXiv:2405.18662.
- Gallegos et al. (2024). *Bias and Fairness in LLMs: A Survey*. *Computational Linguistics*. arXiv:2309.00770.
- Mazeika et al. (2024). *HarmBench*. ICML. arXiv:2402.04249. · Chao et al. (2024). *JailbreakBench*. arXiv:2404.01318.
- SORRY-Bench (2024). arXiv:2406.14598. · Souly et al. (2024). *StrongREJECT*. arXiv:2402.10260.
- Zou et al. (2023). *GCG / Universal Adversarial Attacks*. arXiv:2307.15043. · Chao et al. (2023). *PAIR*. arXiv:2310.08419.
- Mehrotra et al. (2023). *TAP*. arXiv:2312.02119. · Samvelyan et al. (2024). *Rainbow Teaming*. arXiv:2402.16822.
- Scale AI (2024). *Multi-turn Human Jailbreaks (MHJ)*. arXiv:2408.15221. · Russinovich et al. (2024). *Crescendo*. arXiv:2404.01833.
- Anthropic (2024). *Many-shot Jailbreaking*. anthropic.com/research/many-shot-jailbreaking; NeurIPS 2024.
- Andriushchenko et al. (2024). *Simple Adaptive Attacks*. ICLR 2025. arXiv:2404.02151.
- Perez et al. (2022). *Red Teaming LMs with LMs*. arXiv:2202.03286. · Ganguli et al. (2022). *Red Teaming to Reduce Harms*. arXiv:2209.07858.
- Phuong et al. (2024). *Evaluating Frontier Models for Dangerous Capabilities*. arXiv:2403.13793.
- Ren et al. (2024). *Safetywashing*. arXiv:2407.21792. · Bean et al. (2025). *Construct Validity in LLM Benchmarks*. arXiv:2511.04703.
- Li et al. (2024). *LLM-as-a-judge: A Survey*. arXiv:2411.16594. · *A Coin Flip for Safety* (2026). arXiv:2603.06594.

- Singh et al. (2025). *The Leaderboard Illusion*. arXiv:2504.20879. · Chang et al. (2023). *A Survey on Evaluation of LLMs*. arXiv:2307.03109.
- *What AI evaluations can and cannot do* (2024). arXiv:2412.08653. · *Benchmark Data Contamination: A Survey* (2024). arXiv:2406.04244.